

Disproof of Conjecture 00000006289

earthking11

October 6, 2026

A two-element counterexample

Let $C_2 = \{e, s\}$ with $s^2 = e$. Consider its trivial one-dimensional representation

$$\rho(g) = [1] \quad (g \in C_2).$$

It is a representation because $[1][1] = [1]$, so $\rho(gh) = \rho(g)\rho(h)$ for all g, h .

The associated character (matrix trace) is

$$\chi(e) = \text{tr}[1] = 1, \quad \chi(s) = \text{tr}[1] = 1.$$

In particular $s \neq e$ but $\chi(s) \neq 0$. Since C_2 is abelian, its two conjugacy classes are $\{e\}$ and $\{s\}$. Hence the support of χ is both conjugacy classes, i.e. all of C_2 , rather than the identity class.

Proposition 1. *The trace of a finite-group representation need not vanish outside the identity conjugacy class.*

Proof. The trivial character of C_2 has value 1 on the nonidentity element s . □

The identity-only support property does hold for the regular character: its permutation matrices have trace zero away from the identity. The filed statement, however, does not restrict the trace to the regular character, so the trivial character is within scope and disproves it.

The Lean project gives the complete two-element multiplication and support tables and checks them in the kernel.